

Typesetting N'ko Mathematical Equations in LibreOffice

Overview

N'ko is the writing system used primarily for the Manding languages. The script runs completely right-to-left, unlike Arabic where numbers are composed left-to-right. Proper typesetting of N'ko mathematical equations requires a font that supports mirroring of mathematical symbols like XITS Math as well as a type setting program that supports right-to-left formula composition like LibreOffice.

Mathematical Conventions

	Example	Latin	(N'ko) ߞߟߣߊ
<i>Digits</i>		1 2 3 4 5 6 7 8 9 0	ߡ ߢ ߣ ߤ ߥ ߦ ߧ ߨ ߩ ߪ
<i>Letters</i>		a-z/A-Z	ߦߧ-ߩ
<i>Parameter/ Coefficients</i>	$ax + by$	a-d	ߨ-ߩ
<i>Cartesian Coordinates</i>		x, y, z	ߡ ߢ ߣ
<i>Length/Width</i>		l/w	ߡ\ߢ
<i>Radius/Diameter</i>	$d = 2r$	r/d	ߢ\ߡ
<i>Angles</i>	$\cos(\theta)$	$\alpha, \beta, \gamma, \theta, \varphi$	ߡ ߢ ߣ ߤ ߥ ߦ
<i>Functions</i>	$f(x)$	f, g, h	ߢ ߣ ߤ
<i>Decimal Separator</i>	1.25	.	.
<i>Thousands Separator</i>	1,250	,	,
<i>Mean</i>		μ	ߢ
<i>Integer Variables/Indices</i>	$\sum_{i=1}^n i$	i-k	ߡ-ߢ
<i>Fixed Integer/Count/ Probability</i>	$\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$	n	ߡ

Typesetting

LibreOffice supports the mirroring of all mathematical symbols. This handles the majority of cases as long as the formula editor is set to Right-to-Left mode. This is done by looking for the following button.



For example the simply by placing the editor in Right-To-Left mode the following formulas automatically mirror.

LTR	RTL
$\square < \square$	$\square > \square$
$\square \leq \square$	$\square \geq \square$
$\sqrt{\square}$	$\sqrt{\square}$
$\square \in \square$	$\square \ni \square$
$\int \square$	$\square \int$
$\oint \square$	$\square \oint$

Functions with specific names have been added to the user interface. Below is a complete list of those that have been included.

Logarithms/Exponentials

Latin	N'ko
$e^{\square}$	$\square_{\circ}$
$\ln(\square)$	$(\square)_{\text{aɗm}}$
$\exp(\square)$	$(\square)_{\text{ɓɩɩɗ}}$
$\log(\square)$	$(\square)_{\text{ɗɩm}}$

Limits

Latin	N’ko
$\lim \square$	$\square \text{ ፒጦ}$
$\liminf \square$	$\square \text{ ጠፒጦ}$
$\limsup \square$	$\square \text{ ጠፒጦ}$

Trigonometry

Latin	N’ko
$\sin(\square)$	$(\square) \text{ ለኃ}$
$\cos(\square)$	$(\square) \text{ ኃለኃ}$
$\tan(\square)$	$(\square) \text{ ኮለ}$
$\sec(\square)$	$(\square) \text{ ጠፕ}$
$\csc(\square)$	$(\square) \text{ ኃጠፕ}$
$\cot(\square)$	$(\square) \text{ ኃኮለ}$
$\sinh(\square)$	$(\square) \text{ ፕለኃ}$
$\cosh(\square)$	$(\square) \text{ ፕኃለኃ}$
$\tanh(\square)$	$(\square) \text{ ፕኮለ}$
$\coth(\square)$	$(\square) \text{ ፕኃኮለ}$
$\operatorname{sech}(\square)$	$(\square) \text{ ፕጠፕ}$
$\operatorname{csch}(\square)$	$(\square) \text{ ፕኃጠፕ}$
$\arcsin(\square)$	$(\square) \text{ ኃለኃ}$
$\arccos(\square)$	$(\square) \text{ ኃኃለኃ}$
$\arctan(\square)$	$(\square) \text{ ኃኮለ}$

Font Choices

XITS Math includes 5 fonts: XITS Math Regular, XITS Math Bold, XITS Math Italic, XITS Math Unjoined Regular, and XITS Math Unjoined Italic. For typesetting N’ko it is important to distinguish between variables and functions. Therefore variable should use the Unjoined variations. This ensures that two variables next to one another can be read as distinct. Though it is not necessary setting functions a Bold font can help them be more recognizable.

LibreOffice provides the ability to select the font for different elements of a formula. Below is an optimal setup. The bold font not necessary for legible equations but can be used if needed.

The screenshot shows the 'Fonts' dialog box in LibreOffice. It has a title bar with 'Fonts' and a close button. The dialog is divided into two sections: 'Formula Fonts' and 'Custom Fonts'. In the 'Formula Fonts' section, there are five dropdown menus: 'Math:' set to 'XITS Math', 'Variables:' set to 'XITS Math Unjoined, Italic', 'Functions:' set to 'XITS Math, Bold', 'Numbers:' set to 'XITS Math', and 'Text:' set to 'XITS Math'. In the 'Custom Fonts' section, there are three dropdown menus: 'Serif:' set to 'Liberation Serif', 'Sans-serif:' set to 'Liberation Sans', and 'Fixed-width:' set to 'Liberation Mono'. At the bottom, there are five buttons: 'Help', 'Modify' (with a dropdown arrow), 'Default', 'Cancel', and 'OK'.

Sample Formulas

Quadratic Equation:

$$\frac{0 \pm \sqrt{b^2 - 4ac}}{2a} = x$$

Slope Formula:

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y - y_1}{x - x_1}$$

Taylor Series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

L'Hôpital's Rule:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

L'Hôpital's Rule with Trigonometry:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = \frac{\cos 0}{1} = \frac{1}{1} = 1$$